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LAB:	**08**
COURSE TITLE:	**Computer Networks**
COURSE CODE:	**CS-2203**
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LAB 08 – BASIC ROUTER ACCESS AND CLI CONFIGURATION

1. OBJECTIVE

1. Access a router using the Cisco IOS Command Line Interface (CLI)
2. Understand and navigate different Cisco IOS operational modes
3. Configure basic and encrypted passwords for device security
4. View running and startup configuration files
5. Save router configuration to NVRAM

2. THEORY & KEY CONCEPTS

Routers must be configured before deployment. The CLI is the preferred method among network professionals due to its precision and extensive feature access.

Cisco IOS Modes:

Mode	Prompt	Purpose
User EXEC	Router>	Basic monitoring commands
Privileged EXEC	Router#	Advanced commands, debug, config access
Global Configuration	Router(config)#	Device-wide configuration changes

Configuration Files:

RUNNING CONFIG — Active configuration stored in RAM (show running-config)

STARTUP CONFIG — Saved configuration in NVRAM, loaded on reboot (show startup-config)

Password Types:

ENABLE PASSWORD — Basic, weakly protected password

ENABLE SECRET — MD5-encrypted password (preferred; overrides enable password)

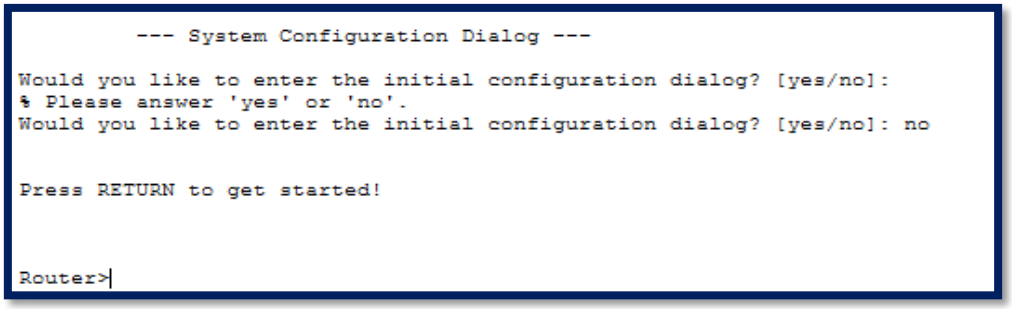
SERVICE PASSWORD-ENCRYPTION — Applies Type 7 encryption to all plaintext passwords

3. SOFTWARE & APPARATUS

- Cisco Packet Tracer
- 1 Router (1941/2911 model), 1 PC
- Console cable (in real lab environment)

4. PROCEDURE & CONFIGURATION COMMANDS

Step 1 — Open Packet Tracer & Access CLI



Place a router in the workspace, click it, open the CLI tab, and press Enter.

STEP 2 — ENTER PRIVILEGED EXEC MODE

Router> enable
Router#

```
Router>enable
Router#
```

STEP 3 — ENTER GLOBAL CONFIGURATION MODE

Router# configure terminal
Router(config)#

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
```

STEP 4 — SET ENABLE PASSWORD

Router(config)# enable password cisco123

```
Router(config)#enable password cisco123
Router(config)#
```

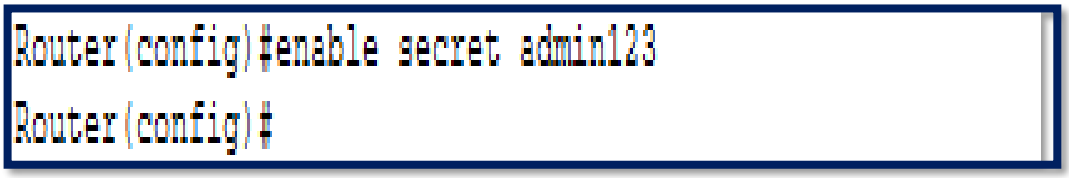
STEP 5 — ENABLE PASSWORD ENCRYPTION

Router(config)# service password-encryption

```
Router(config)#service password-encryption
Router(config)#
```

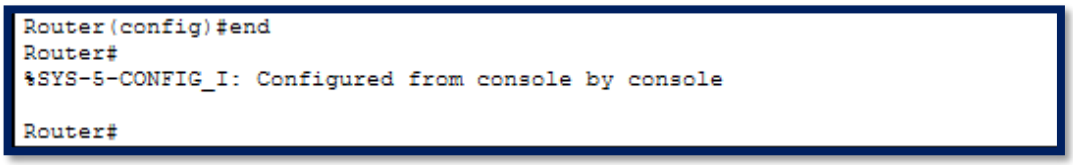
STEP 6 — SET ENABLE SECRET (STRONGER)

Router(config)# enable secret admin123



STEP 7 — EXIT CONFIGURATION MODE

Router(config)# end
Router#



STEP 8 — VIEW RUNNING CONFIGURATION

Router# show running-config



STEP 9 — VIEW STARTUP CONFIGURATION

Router# show startup-config

```
Router#show startup-config
startup-config is not present
Router#
```

STEP 10 — SAVE CONFIGURATION TO NVRAM

Router# copy running-config startup-config

```
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

5. COMMAND SUMMARY TABLE

Command	Mode	Purpose
enable	User EXEC	Enter Privileged EXEC mode
configure terminal	Privileged EXEC	Enter Global Configuration mode
enable password <pass>	Global Config	Set basic privileged mode password
service password-encryption	Global Config	Encrypt all plaintext passwords
enable secret <pass>	Global Config	Set MD5-encrypted privileged password
show running-config	Privileged EXEC	Display active (RAM) configuration
show startup-config	Privileged EXEC	Display saved (NVRAM) configuration
copy running-config startup-config	Privileged EXEC	Save config to NVRAM
end / Ctrl+Z	Any Config mode	Return to Privileged EXEC

6. OBSERVATIONS & RESULTS

- Successfully navigated all IOS modes: User EXEC → Privileged EXEC → Global Config
- `enable secret` takes precedence over `enable password` when both are set
- After applying `service password-encryption`, passwords appeared encrypted (Type 7) in the running config
- Running config showed `693 bytes`; startup config showed "not present" before saving
- After `copy running-config startup-config`, the startup config reflected all changes correctly

7. DISCUSSION & LEARNING OUTCOMES

This lab established foundational skills in Cisco device management. Key takeaways:

- CLI mode awareness is critical — incorrect mode leads to command failures
- The distinction between running and startup config is essential; unsaved changes are lost on reboot
- `enable secret` should always be preferred over `enable password` for stronger security
- Packet Tracer accurately simulates real router behavior, making these skills directly transferable to physical hardware and CCNA-level networking

8. CONCLUSION

All lab objectives were successfully achieved. The router was configured via CLI with proper mode navigation, password security (both basic and encrypted), and configuration management.